

Exercise 1 **Stability and Invariant set**

Consider the uncertain one-dimensional system

$$x(k+1) = x(k) + u(k) + w(k)$$

with input constraints $u \in \mathcal{U} = \{u \mid -10 \leq u \leq 10\}$ and a persistent disturbance that lies in the set $w \in \mathcal{W} = \{w \mid |w| \leq 1\}$

Your goal is to design an open-loop robust MPC control law.

1. Compute the maximum robust invariant set $\mathcal{X}_f$ for this system under the control law $u(k) = -x(k)$.
2. Use the set $\mathcal{X}_f$ you computed in the last part to compute a terminal set for an open-loop robust MPC controller with a horizon of 2.
3. Write the matrices of the QP problem associated with the open-loop robust MPC with a horizon of two, a quadratic stage cost $l(x_i, u_i) = 3x_i^T x_i + u_i^T u_i$, and a terminal cost of $5x_2^T x_2$.
4. How long can the horizon N be before the feasible set of the open-loop robust MPC problem is empty?

Exercise 2 **Pre-stabilizing control law**

Consider the uncertain one-dimensional system

$$x(k+1) = 2x(k) + u(k) + w(k)$$

with input constraints $u \in \mathcal{U} := \{u \mid |u| \leq 1\}$ and a disturbance w acting on the system which lies in the bounded set $w \in \mathcal{W} := \{w \mid |w| \leq \frac{1}{2}\}$.

Over the prediction horizon, we propose to utilize the following control law

$$u_i = Kx_i + v_i$$

where $K = -1.5$ and we will optimize over the offset v_i .

- 1) Give the set of states and offsets (x, v) for which the control law $u(k) = Kx(k) + v(k)$ satisfies the system constraints.
- 2) Given that you apply the input sequence $\{v_0, v_1, v_2\}$ starting from the current state x_0 , compute the set of states that the system may be in at times $t = 0, 1, 2, 3$.
- 3) If you want to guarantee that at time $t = 3$, the condition $x_3 \leq 3$ is met, what constraints must hold for offsets $\{v_0, v_1, v_2\}$?